

1. MOTIVATION

Vietnam recorded **7,004** occupational accidents and 658 fatalities in 2025. Human negligence remains a key factor, with **13.75%** of cases involving safety procedure violations and **6.25%** due to failure to use Personal Protective Equipment (PPE) [1]. Meanwhile traditional CCTV monitoring by security personnel lead to fatigue and missed short-duration violations. Cloud-based systems also face high latency (500–1000 ms), heavy bandwidth usage (4–8 Mbps per camera), and significant privacy risks due to continuous raw video transmission.



Figure 1: PPE violated construction site

Related Work	Research Gap
YOLO-based PPE detection with attention modules [2]	Object-focused detection without hazard context.
Edge computing for construction monitoring [3,4]	Bandwidth savings mainly reported as theoretical estimates.
Frame-by-frame vision monitoring [5]	High false-positive rate caused by transient detections.
Model compression for edge AI [6]	Limited validation in real-world deployment scenarios.

Table 1: Related works

CONTRIBUTIONS:

- Context-Aware Alerting:** Combines worker location with PPE status to generate zone-aware WARNING/CRITICAL alerts.
- Bandwidth Efficiency:** Empirically reduced upstream traffic by 99.93% (1.35 MB/hour).
- Deployment Benchmark:** Validated the ONNX FP32 deployment pipeline, achieving 131 FPS in an edge-oriented GPU proxy environment.
- Temporal Smoothing:** Five-frame confirmation reduced false alarms by 94.6% while maintaining 167 ms alert latency.

3. DATASET

The **Ultralytics Construction-PPE** dataset was used as the base dataset. Two additional public datasets were merged through class remapping to increase training samples and improve class balance.

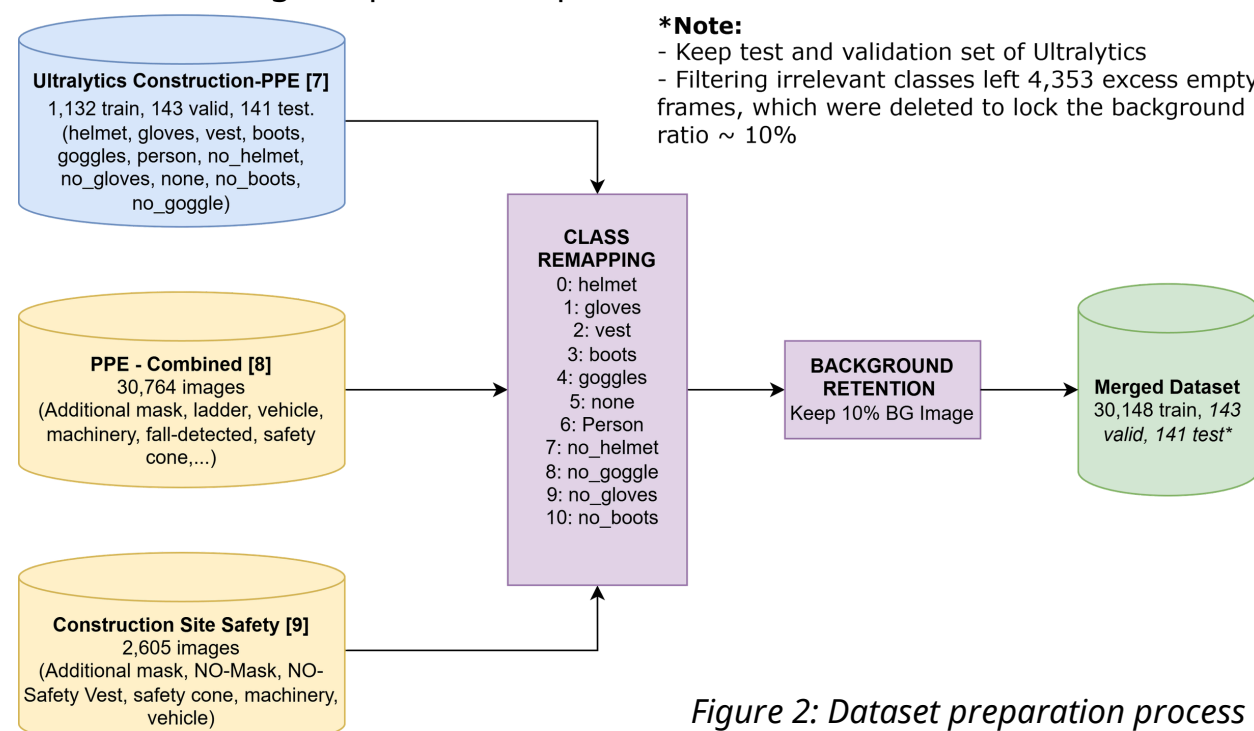


Figure 2: Dataset preparation process

4. METHODOLOGY

a) Overall method

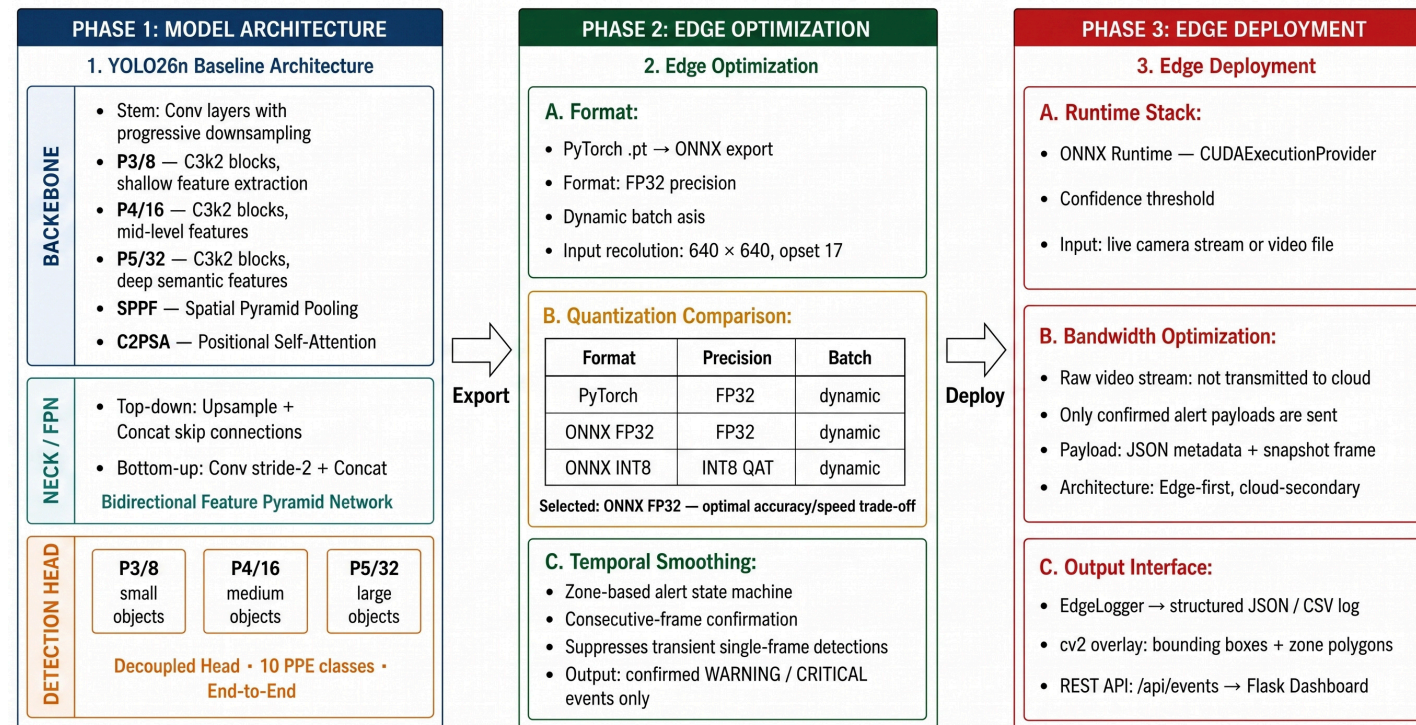


Figure 2: Overall method

b) Workflow

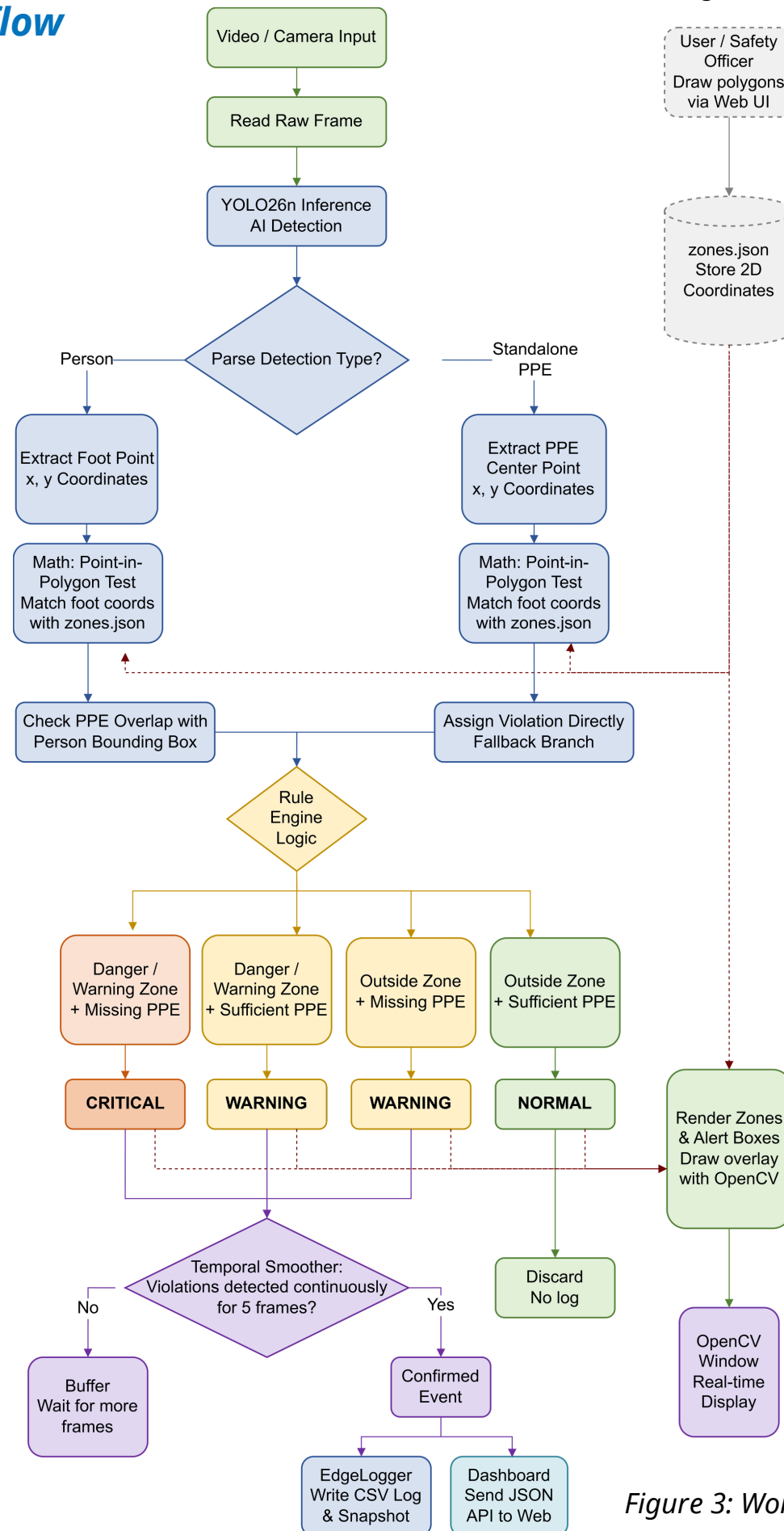


Figure 3: Workflow

5. RESULTS

Model Format	Model Size	Latency (ms/img)	Throughput (FPS)	Overall mAP@0.5	Violation mAP@0.5
ONNX FP32 (Dynamic B=8)	10.3 MB	7.61	131	0.749	0.671
ONNX FP16 (Dynamic B=4)	5.7 MB	9.05	110	0.750	0.671

Table 2: ONNX Quantization Results

Optimization Target	Baseline State	Edge-Optimized Result
Alert Reliability (baseline_yolo26n ONNX FP32)	124 alerts (No temporal filter)	6 alerts (N = 5 frames)
Network Efficiency	1.897 GB/hr (Raw video)	1.35 MB/hr (Logs + Snapshots)*

*Note: Network efficiency assumes 1 event/min.

Table 3: Edge Optimization Result

mAP@0.5	Edge-Optimized Efficiency	End-to-End Pipeline
Violated classes: no_helmet 0.871 no_goggle 0.877 no_gloves 0.810 (yolo26n_base_onnx_f32 B=8)	Reduce 99.93% Bandwidth Reduce 94.6% False positives Achieve 131 FPS Edge throughput*	Context-Aware Logic: Worker position + PPE status Live Dashboard: Real-time telemetry, snapshot logs, and zone mapping.



Figure 4: Snapshot of a violated scene

7. REFERENCES

- [1] M. Hien, "Tình hình tai nạn lao động năm 2025: Giảm về số vụ nhưng vẫn tiềm ẩn nhiều rủi ro," Tạp chí Tổ chức nhà nước và Lao động - Bộ Nội vụ, Mar. 2026.
- [2] Z. Wu et al., "Advancing construction safety: YOLOv8-CGS helmet detection model," PLOS One, vol. 20, no. 5, p. e0321713, May 2025.
- [3] X. Xiao et al., "Exploring Edge Computing for Sustainable CV-Based Worker Detection in Construction Site Monitoring: Performance and Feasibility Analysis," Buildings, vol. 14, no. 8, p. 2299, Jul. 2024.
- [4] M. Gugssa et al., "Enhancing the Time Efficiency of Personal Protective Equipment (PPE) Detection in Real Implementations Using Edge Computing," in Computing in Civil Engineering 2023, Jan. 2024, pp. 532–540.
- [5] S. JoonOh et al., "Computer vision techniques for construction safety and health monitoring," vol. 29, pp. 239–251, 2015.
- [6] B. Tong et al., "A deep learning-based algorithm for the detection of personal protective equipment," PLOS One, vol. 20, no. 5, p. e0322115, May 2025.
- [7] M. Davi et al., "Construction-PPE: Personal Protective Equipment Detection Dataset," Ultralytics, Jan. 2025.
- [8] Roboflow Universe Projects, "Personal Protective Equipment - Combined Model Dataset," Roboflow Universe, Jun. 2025.
- [9] Roboflow Universe Projects, "Construction Site Safety Dataset," Roboflow Universe, Jan. 2026.

6. CONCLUSION

- Successfully deployed an Edge AI pipeline for real-time PPE monitoring, eliminating cloud-related latency, bandwidth, and privacy issues.
- The optimized baseline_yolo26n model (ONNX FP32) achieves 131 FPS at 2.93 ms/frame, enabling concurrent multi-camera processing.
- A 5-frame temporal smoothing algorithm reduces false alarms by 94.6%. The edge-first architecture reduces bandwidth consumption by 99.93% (1.35 MB/hour).
- Future Work:** Enhance no_boots detection, integrate BIM for dynamic safety zones, and apply multi-modal sensor fusion for poor weather conditions.